13.4 Parent Functions

Lesson Introduction

Benchmark	MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input-output table, determine a function type that could represent it.	Learning Targets	• I can match multiple representations of the parent function to its table of values. • I can determine a function type that could represent a transformation of a parent function.

	Building On	Working Toward
Benchmark Coherence	MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.	MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input-output table, determine a function type that could represent it. Algebra 2 goes beyond the clarifications.

Essential Questions	• How can I match a table of values to different representations of its parent function? • How can I determine if a function could represent a transformation of a parent function?
Lesson Narrative	In this lesson, students will engage in a review of parent functions and transformations of them. Students will begin by engaging in a parent function card sort activity where they are provided with a table of values and will match them to the equation, graph, and key features that they represent. They will then transition into exploring parent function transformations by providing multiple representations and then matching them to the equation of the parent function.

Component Summary	13.4.1 Warm-Up (~5 min.)	Notice and Wonder activity with function graphs *(SE p. 254)*	13.4.3 Exploration (15-20 min.)	Parent function transformations table match activity *(SE p. 255)*
	13.4.2 Activity (15-20 min.)	Parent functions card sort activity *(SE p. 254)*	13.4.4 Wrap-Up (5 min.)	Learning target self-reflection *(SE p. 257)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A	• Parent function card sort activity cards	• (Optional) If making the Exploration a gallery walk, chart paper will be needed.

<table>
<tr>
<td rowspan="4">Teacher Tips</td>
<td>Common Misconceptions</td>
<td>Things to Consider</td>
</tr>
<tr>
<td>

Students may not recall how to identify specific parent functions.
Students may misrepresent data in a specific form (i.e., table, graph, or equation).
Students may be unable to identify a function type when expressed in a specific form.

</td>
<td>

The timing of this lesson is critical to unfolding successfully. The problems are laid out as a review of what they’ve already learned in the course. If students are struggling, it can easily cause the timing to slip away from you during the lesson. Consider using a timer to keep things on track.

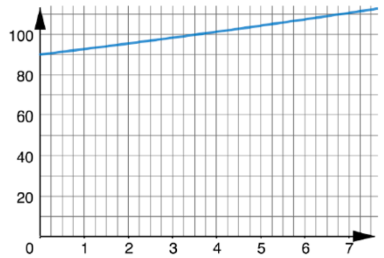
</td>
</tr>
<tr>
<td>Literacy and Language Routines</td>
<td>Mathematical Thinking and Reasoning (MTR) Standard</td>
</tr>
<tr>
<td>Notice and Wonder
13.4.1 Warm-Up provides students with an opportunity to engage in a Notice and Wonder routine using the graphs of a linear and an exponential function.</td>
<td>MA.K12.MTR.2.1: Multiple Representations
13.4.2 Activity provides students with the table of values from multiple functions and has them match it with its respective equation, graph, and key features. The various functions in this activity are presented in multiple ways. Students will express connections between these concepts and their representations.</td>
</tr>
<tr>
<td>Differentiate Instruction</td>
<td colspan="2">For 13.4.3 Exploration, consider adding physical movement and student autonomy to enhance the activity. This can be done by converting this activity into a gallery walk. To do this, the problems from the book can be printed or put onto chart paper and spread throughout the room. Students will get up and walk to a poster and justify how they know the parent function of the representation. They will complete this for each poster in the room.</td>
</tr>
<tr>
<td colspan="3">Lesson Notes:</td>
</tr>
</table>

13.4.1 Warm-Up: Notice and Wonder

Component Overview: *Students will engage in a Notice and Wonder protocol with the graph of a linear and exponential function.*

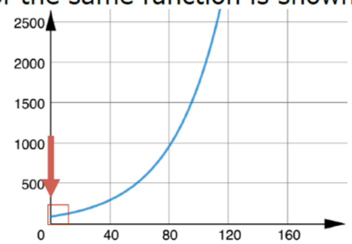
13.4.1 Warm-Up: Notice and Wonder

1. A graph of a function is shown.



What information might help you decide whether the graph represents a linear or an exponential function?
Slope or rate of change of the graph.

2. The graph of the same function is shown.



Explain why the x-values used for the first graph, shown in the red box, were misleading.
It appeared to be a straight line. The graph shows a curve.

For this activity, consider leveraging a Think-Pair-Share routine. Allow students a couple of minutes to try questions 1 and 2 on their own and then pair up with a student to discuss and compare notes. Set up an opportunity for student pairs to share their thinking for each question so that others can hear and engage in the conversation.

13.4.2 Activity: Parent Functions Card Sort

***Component Overview**: Students will be provided with eight tables of values that represent eight different functions. They will work with a partner or small group to match the table of values to its corresponding equation, graph, and key features.*

13.4.2 Activity: Parent Functions Card Sort

Sort the cards to match the equation, graph, and key features to the table of values of a parent function. Write the letter that corresponds to each in the table.

<table>
<tr><th>Table of Values</th><th>Equation</th><th>Graph</th><th>Key Features</th></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-8</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>8</td></tr></table></td><td>D</td><td>W</td><td>K or T</td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>2</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr></table></td><td>B</td><td>G</td><td>Z</td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr></table></td><td>V</td><td>U</td><td>N</td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>$\frac{1}{4}$</td></tr><tr><td>-1</td><td>$\frac{1}{2}$</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr></table></td><td>R</td><td>Y</td><td>H</td></tr>
</table>

For this activity, students will write down the letter of the card that matches the table of values in the book. To best prepare for this activity, you will need the cards prepared to create a smooth transition with this activity.

What are some of the strategies that you used when matching the cards to the tables?

13.4.2 Activity: Parent Functions Card Sort (continued)

<table>
<tr><th>Table of Values</th><th>Equation</th><th>Graph</th><th>Key Features</th></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>4</td><td>2</td></tr><tr><td>9</td><td>3</td></tr><tr><td>16</td><td>4</td></tr></table></td><td>X</td><td>A</td><td>M</td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>−2</td><td>4</td></tr><tr><td>−1</td><td>2</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>$\frac{1}{2}$</td></tr><tr><td>2</td><td>$\frac{1}{4}$</td></tr></table></td><td>Q</td><td>E</td><td>O</td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>−2</td><td>4</td></tr><tr><td>−1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr></table></td><td>C</td><td>P</td><td>S</td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>−8</td><td>−2</td></tr><tr><td>−1</td><td>−1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>8</td><td>2</td></tr></table></td><td>F</td><td>L</td><td>T or K</td></tr>
</table>

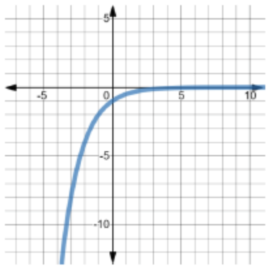
If technology is available, consider going digital with this card sort activity. Use a technology tool such as Desmos to create a card sort activity. This could increase engagement and provide easier-to-read data on student performance with this task.

13.4.3 Exploration: Transformations of Parent Functions

Component Overview: *Students will be provided seven different representations of different functions that have gone through a transformation. Students will match the representation with the equation of the parent function.*

13.4.3 Exploration: Transformations of Parent Functions

Knowing the basic shape of a parent function helps to determine a transformation of the parent function. Shown in the table are representations of transformations. Use the information from the card sort to help determine which family of functions each transformation is part of.

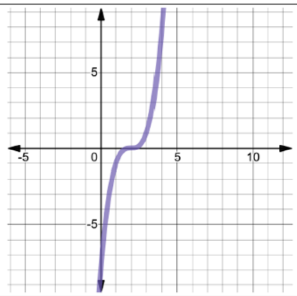
<table>
<tr><th>Representation of Transformation</th><th>Equation of Parent Function</th></tr>
<tr><td>
</td><td>○ $y = x$
○ $y = x^2$
○ $y = x^3$
○ $y = \sqrt{x}$
○ $y = |x|$
○ $y = 2^x$
○ $y = \frac{1^x}{2}$</td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>3</td></tr><tr><td>-1</td><td>4</td></tr><tr><td>0</td><td>5</td></tr><tr><td>1</td><td>6</td></tr><tr><td>2</td><td>7</td></tr></table></td><td>○ $y = x$
○ $y = x^2$
○ $y = x^3$
○ $y = \sqrt{x}$
○ $y = |x|$
○ $y = 2^x$
○ $y = \frac{1^x}{2}$</td></tr>
<tr><td>$y = |x + 5|$</td><td>○ $y = x$
○ $y = x^2$
○ $y = x^3$
○ $y = \sqrt{x}$
○ $y = |x|$
○ $y = 2^x$
○ $y = \frac{1^x}{2}$</td></tr>
</table>

Consider adding physical movement and student autonomy to enhance the activity. This can be done by converting this activity into a gallery walk. To do this, the problems from the book can be printed or put onto chart paper and spread throughout the room. Students will get up and walk to a poster and justify how they know the parent function of the representation. They will complete this for each poster in the room.

13.4.3 Exploration: Transformations of Parent Functions (continued)

x	-1	0	1	2	3
y	$\frac{1}{4}$	$\frac{1}{2}$	0	1	4

- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$



- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

x	y
-2	6
-1	3
0	0
1	3
2	6

- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

$$y = \left(\frac{1}{2}\right)^x - 5$$

- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

When the activity is completed, consider enacting a polling protocol. Display the representation on a screen or central location. Then call out one of the functions and have students raise their hands if they think the function represents the information. Consider taking counts to see how many students were correct and how many were not to get an idea for potential misconceptions and how prevalent they may be.

13.4.4 Wrap-Up: Reflection

***Component Overview**: Students will complete a self-reflection where they rate themselves on the learning targets from this lesson.*

13.4.4 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about each learning target.

Answers will vary.

1. I can match multiple representations of the parent function to its table of values.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can determine a function type that could represent a transformation of a parent function.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.